

HOTEL ANALYTICS PROJECT

Predicting Booking Cancellations

A Machine Learning and Neural Network Comparative Analysis of "HotelData.xlsx"

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Project Scope & Objectives

Primary Goal: Build a highly accurate predictive pipeline to determine whether a hotel booking is likely to be canceled, based on historical properties in "HotelData.xlsx".

- Reduce lost revenue caused by last-minute reservation cancellations.
- Compare traditional tuned Machine Learning against Deep Neural Networks.
- Formulate optimal feature selection and robust handling of dataset class imbalances.
- Utilize model metrics (Recall & Accuracy) to evaluate and validate findings.



Exploratory Data Analysis



Class Imbalance

The target variable (`booking_status`) is heavily skewed. Cancellations represent only about 28% of the data, requiring careful handling before modeling.



Lead Time Influence

Longer lead times correlate strongly with cancellations. Guests booking far in advance are statistically more likely to alter or cancel their stays.



Pricing Metrics

Average room price plays a distinct role, especially when evaluated relative to the composition of guests on the reservation.

Custom Feature Engineering



total_nights

Calculated as the sum of weekend nights and weeknights. Helps the model understand overall stay duration and its relationship to cancellations.



total_guests

Derived by adding adults and children together to establish the true occupancy size of each booking room.



Ratios & Dates

Engineered price_per_guest, weekend_ratio, and extracted arrival_day_of_week after compiling full arrival dates.

Data Preprocessing Pipeline



Date Standardization: Combined year, month, and day into full datetime objects. Dropped rows with parsing anomalies (e.g., February invalid dates) to maintain clean records.



Log Transformations: Handled heavy right-skewness on continuous variables (`lead_time`, `avg_price_per_room`, `price_per_guest`) using logarithmic transformations.



Encoding Categoricals: Converted target variable `booking_status` to binary integers (1 for Canceled, 0 for Not Canceled) and dummy-encoded other categorical features.

Handling Class Imbalance

29,622

Balanced Sample Size

SMOTE Oversampling

Using the original unbalanced training dataset often forces models to over-predict the majority class (Not Canceled).

We applied Synthetic Minority Over-sampling Technique (SMOTE) to the training split, generating synthetic cancellations to balance both classes perfectly at a 50/50 ratio. Features were subsequently scaled using MinMaxScaler.

Model Exploration & Showdown

Comparing tuned Random Forest against a Deep Neural Network on
"HotelData.xlsx"

Model 1: Optimized Random Forest

Hyperparameter Tuning

Tuned via RandomizedSearchCV across multiple iterations to locate optimal boundaries and avoid overfitting. Scoring was optimized on the F1 Metric to capture precision and recall balance.

Optimal Parameters

n_estimators: 200 trees
max_depth: 15 levels
min_samples_split: 10
min_samples_leaf: 4
max_features: 80% (0.8)

Model 2: Deep Neural Network

Robust MLP Architecture

Constructed a feedforward neural network with 4 hidden layers: **512** → **256** → **128** → **64** nodes, ending with a single sigmoid-activated output node for binary probability classification.

Regularization & Control

Implemented aggressive Dropout (0.4 to 0.2) after each dense layer to manage neural redundancy. Leveraged Adam Optimizer, binary crossentropy loss, and Early Stopping with 15-epoch patience.

Comparative Performance Table

Prediction Model	Test Accuracy	Test Recall (Canceled)	Testing Support
Random Forest (Tuned)	82.58%	76.05%	5,156 bookings
Deep Neural Network	79.27%	70.20%	5,156 bookings

Evaluated strictly on held-out testing data. Random Forest outshines the Deep Neural Network across all primary metrics on this tabular dataset.

Visual Performance Metrics

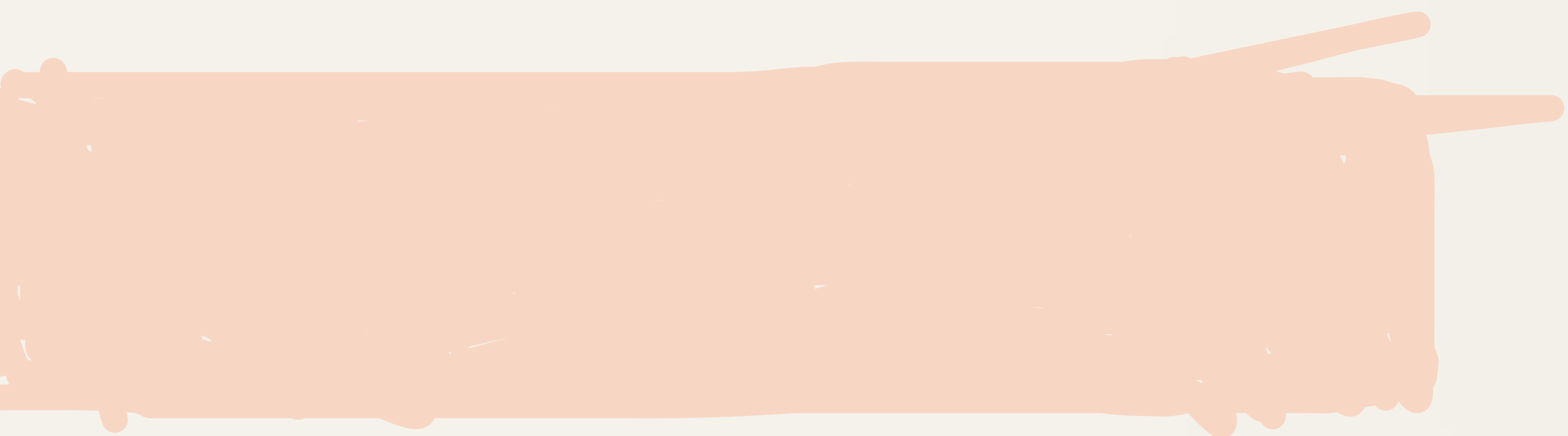


The Random Forest model displays a notable advantage, showing a ~3.3% lead in overall accuracy and nearly a 6% boost in catching booking cancellations (Recall).

Key Takeaways

Random Forest proved superior at mapping tabular features in "HotelData.xlsx". SMOTE oversampling significantly elevated cancellation recall across both models.

Thank you for your attention | Questions & Discussion



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